2.5 Intermolecular forces

Students will be assessed on their ability to:

a demonstrate an understanding of the nature of intermolecular forces resulting from interactions between permanent dipoles, instantaneous dipoles and induced dipoles (London forces) and from the formation of hydrogen bonds

b relate the physical properties of materials to the types of intermolecular force present, eg:

- i the trends in boiling and melting temperatures of alkanes with increasing chain length
- ii the effect of branching in the carbon chain on the boiling and melting temperatures of alkanes
- iii the relatively low volatility (higher boiling temperatures) of alcohols compared to alkanes with a similar number of electrons
- iv the trends in boiling temperatures of the hydrogen halides HF to HI

c carry out experiments to study the solubility of simple molecules in different solvents